

MUSEO DE BALER · BALER, AURORA

Operating the System

How the Museo de Baler system is used, screen by screen: admitting visitors at the door, maintaining the collection, guiding visitors through the museum on their own phones, producing the reports the LGU files, and recovering when something goes wrong.

With the system's stack and code inventory, for readers who need to know what it runs on.

Deployed at <https://museodebaler.com>

Conventions

- **“The panel”** means the website staff sign in to on a computer or tablet. **“The app”** means what a visitor opens on their own phone. They are the same software and the same database, reached by two different doors.
- **Monospaced text** is something typed exactly as shown: a command, a file path, a configuration key, or a URL.
- **Bold** inside an instruction is a button or a menu item on the screen.
- Figures are screenshots of the running system, not mock-ups. Where this manual and the screen disagree, the screen is correct and this document is out of date.

Document control

Item	Value
Document	Museo de Baler — Operating the System
Version	1.0
Date	29 September 2026
Software stack	Laravel 12, PHP 8.3 (8.2 minimum), MySQL 8, vanilla-JavaScript PWA
Production address	https://museodebaler.com

One idea to understand before anything else. A visitor's phone receives an access token when they register, but exhibit content is released only after the front desk has collected the admission fee or sighted a Baler resident's identification. That decision is made on the server, on every single request. The app's “waiting for the desk” screen is a rendering of the server's answer, never the check itself. Almost everything in this document follows from that sentence.

A1 · What the system is	4
A1.1 The addresses	5
A1.2 What replaces what	5
A2 · Signing in, and who can do what	6
A2.1 The two roles	6
A3 · The visitor's journey on a phone	8
A3.1 The sequence	8
A3.2 Settings the visitor controls	10
A3.3 What the visitor keeps when the signal drops	10
A4 · The front desk	11
A4.1 Three ways a visitor gets in	11
A4.2 Clearing a visitor	11
A4.3 Groups: schools, families, tour agencies	12
A4.4 Refunds and corrections	13
A4.5 End of day	13
A5 · Exhibits: add, edit, archive	14
A5.1 Adding an exhibit	14
A5.2 Editing an exhibit	15
A5.3 Archive and restore	16
A6 · Translations and the audio guide	17
A6.1 Method A: let the AI draft them	17
A6.2 Method B: type a translation and upload a recording	18
A7 · QR codes and the entrance poster	19
A8 · The museum map and the storyline	21
A9 · Museum info: hours, fee, halls, the geofence	22
A9.1 The fields	22
A9.2 The geofence	23
A10 · Camera recognition	24
A11 · Guided tours and visitor feedback	25
A11.1 Tours	25
A11.2 Feedback and the ARTA survey	25
A12 · The Tourism office's screens	27
A12.1 Staff accounts	27
A12.2 Staff attendance	27
A12.3 Attendance corrections	28
A12.4 The visitor survey	29
A12.5 The audit trail	30
A13 · Reports and exports	32
A13.1 The six reports and their formats	32
A13.2 The audit trail is Tourism-only	33
A14 · Troubleshooting	34
A14.1 When a visitor asks for help	34
A14.2 When the panel shows an error	35
A15 · Routine maintenance	36
A15.1 Weekly, whoever opens the museum	36
A15.2 Quarterly, the Tourism office with the maintainer	36
B1 · Stack and inventory	37
B1.1 Platform	37
B1.2 Code inventory	37
B1.3 What is authored and what is generated	38

A1 · What the system is

Museo de Baler runs as one application with three faces. They share one database, so a fee collected at the desk is visible on the visitor's phone a second later, and a scan made in a hall appears in the evening's report.

Face	Who uses it, and where	What it does
The staff panel	Museum staff and the Municipal Tourism Office, on an office computer or the counter tablet	The front desk, the exhibit collection, translations and narration, QR labels, the museum map, guided tours, staff attendance, visitor feedback, and every report
The visitor app	Visitors, on their own phones	Register, wait for the desk to clear them, then scan or photograph exhibits, read and listen in their chosen language, follow the storyline or roam freely, and leave feedback
The API	Nobody directly — the app talks to it	Serves the app its data as JSON and enforces the admission gate on every request

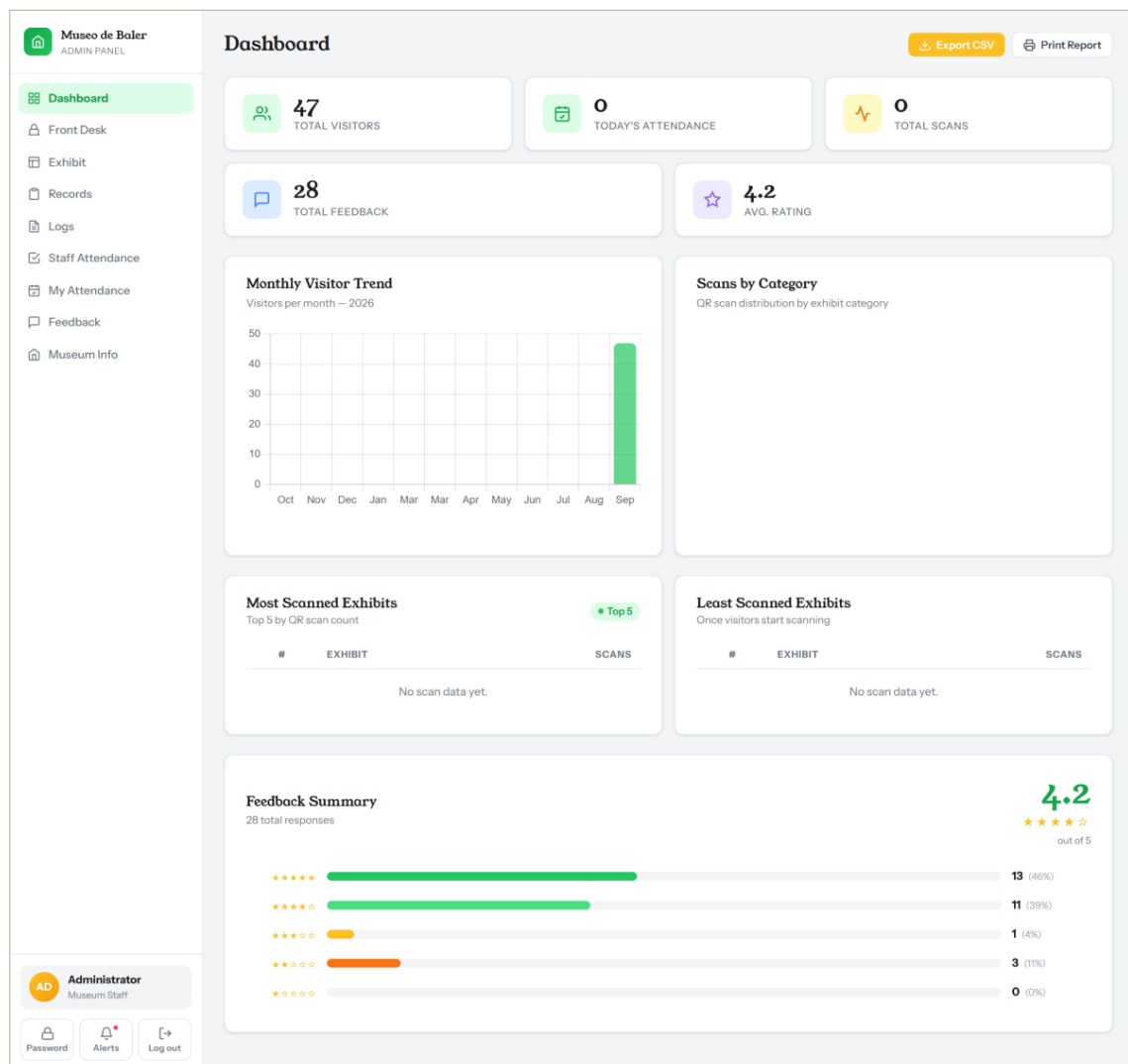


Figure A1.1 — The dashboard, which is where museum staff land after signing in: the day's counts, the visitor chart, and recent activity.

A1.1 The addresses

Who	Where they go
Visitors	https://museodebaler.com — the bare domain opens the visitor app
Visitors, from a display case	https://museodebaler.com/visitor/?scan=EXH-2026-001 — what a QR label encodes
All staff, both roles	https://museodebaler.com/login

There is no separate Tourism website. Both roles sign in at the same address; what each one sees afterwards is decided by their role, so the head of tourism never sees the exhibit editor and museum staff never see account creation or the audit trail.

A1.2 What replaces what

- The **paper logbook** at the entrance becomes the Front Desk screen, which also prices admission, handles a whole party with one payment, and prints the day for the cash count.
- **Printed exhibit labels in one language** become an exhibit record carrying a title, description, fun facts and spoken narration in English, Filipino and Spanish.
- The **paper daily time record** becomes a clock-in by phone from inside the museum grounds, with a monthly DTR the Tourism office prints.
- The **ARTA Client Satisfaction survey** on paper becomes a sheet in the app, with the CSM figures and the CSV the office files with the LGU produced from the answers.

A2 · Signing in, and who can do what

Open the panel's address in a browser **on a computer or tablet**. The panel is laid out for a screen that size; on a phone it deliberately narrows to clocking in and changing your own password, which are the only two things that need one.

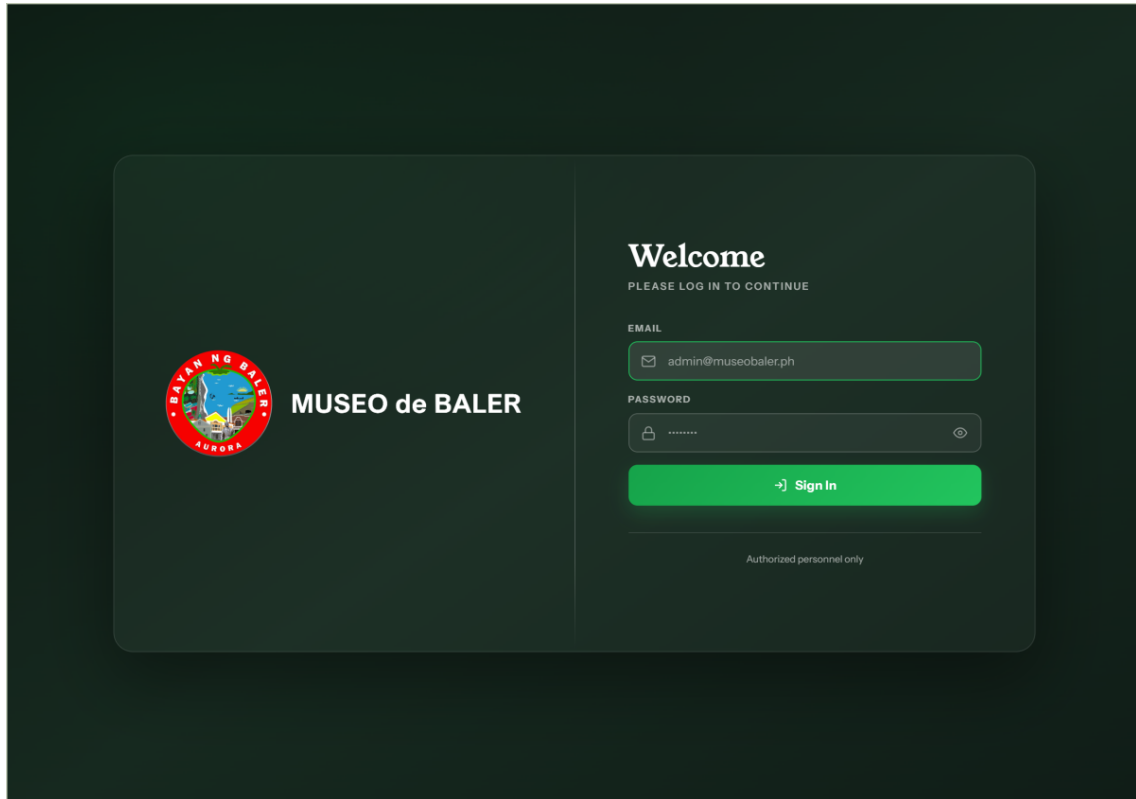


Figure A2.1 — The staff sign-in screen. The same address serves both roles.

1. Enter your email address and the password you were issued.
2. **The first time, you must choose a new password.** It must be at least 12 characters with upper and lower case, a number and a symbol, and may not contain your name or the museum's name.
3. After repeated wrong attempts the system pauses sign-in briefly. If you have forgotten your password, the Tourism office issues a new one (section A12).

A2.1 The two roles

There are exactly two kinds of staff account. The split is deliberate, and it is enforced by the server on every request rather than by hiding menu items.

Can do	Administrator (museum staff)	Tourism Head (Municipal Tourism Office)
Front desk: register visitors and groups, collect fees, verify IDs	Yes	—
Exhibits, translations, narration, QR codes, gallery images	Yes	—
The museum map and the storyline path	Yes	—
Museum info: hours, admission fee, halls, the geofence	Yes	—

Can do	Administrator (museum staff)	Tourism Head (Municipal Tourism Office)
Camera recognition: training photos and training the model	Yes	—
Guided tours	Yes	—
Clock in and out; file an attendance correction	Yes	—
Visitor records, feedback, activity log, staff attendance sheet	Yes	Yes
Reports: logbook, DTR, visitors, exhibits, feedback	Yes	Yes
Create staff accounts and reset their passwords	—	Yes
Staff work schedules	—	Yes
Approve or reject attendance corrections	—	Yes
Edit the visitor survey question bank	—	Yes
Export the audit trail	—	Yes

Why the split is drawn here. The museum is staffed by the LGU tourism office. If museum staff could create accounts, the tier being overseen would be minting its own overseers. Work schedules sit with Tourism because a schedule decides who counts as late. Approvals sit with Tourism because a hand-written attendance row needs a second signature from someone who did not ask for it — so no one person can both file a correction and approve it, and nobody can file one for themselves.

Sign out from the menu under your name whenever you leave a shared computer. Changing your password signs out every other session on your account.

A3 · The visitor's journey on a phone

This is the flow a panel will most likely ask to see end to end. Nothing here is installed from an app store: the visitor app is a website that installs itself from the browser.

A3.1 The sequence

#	Screen	What happens
1	Register or Sign in	The visitor scans the entrance poster's QR code, or types the address. They register with an email address and a password of their choosing, or sign in if they have visited before.
2	Visitor information	Name, age, sex, visitor type (local resident, domestic tourist, foreign visitor) and where they are from. A Baler resident picks their barangay. Someone joining a party already registered at the desk enters the party's six-letter code here.
3	Waiting for the desk	The screen says either <i>Hand your admission fee to the staff at the desk or Show your ID at the entrance desk to claim free entry</i> . No exhibit content exists on the phone yet. The screen unlocks by itself the moment staff act.
4	How would you like to explore?	Two modes. Follow the Storyline walks the curated narrative order and, after each scan, highlights the next stop on the map. Explore Freely allows any exhibit in any order. The choice can be changed later in Settings.
5	Home	The starting point: the scanner, the exhibit list, the map, and the visitor's own progress.
6	Scan	Either point the camera at the QR label on the display case, or point it at the object itself and let the app recognise it (section A10). If neither works, the exhibit code printed under the QR square can be typed instead.
7	Exhibit	Three tabs: Overview (the description, with the narration player), Fun Facts , and Gallery . Narration can start automatically on scan and plays at an adjustable speed. Where no recording exists for the chosen language, the phone reads the description aloud instead and says so.
8	Map	Both floors, with a pin per exhibit, the ones already found marked, and in storyline mode the path from the current stop to the next.
9	Profile	What the visitor has scanned, what they bookmarked, and which halls they have covered.
10	Feedback	The ARTA Client Satisfaction sheet. A visitor who was on a guided tour is additionally asked about their guide; nobody else sees that question.

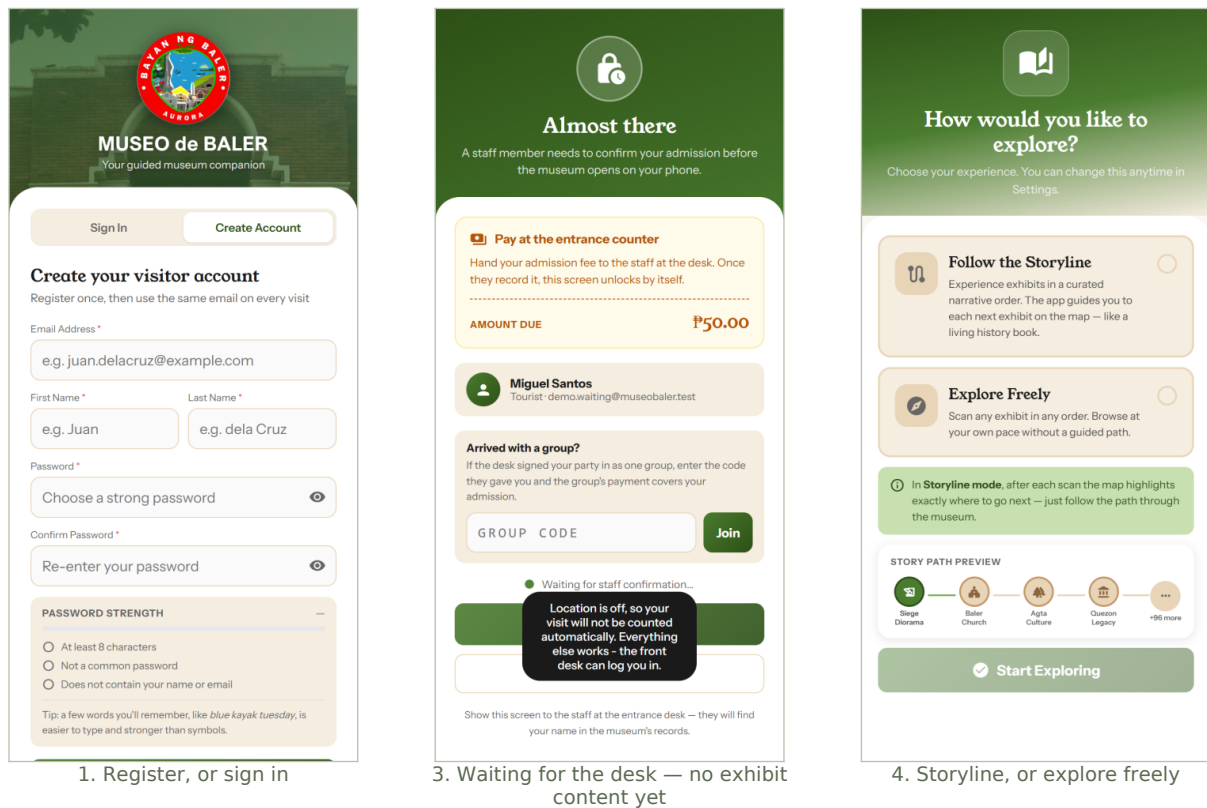


Figure A3.1 — Getting in. The middle screen is the admission gate as the visitor sees it; it unlocks by itself the moment the desk takes the fee.

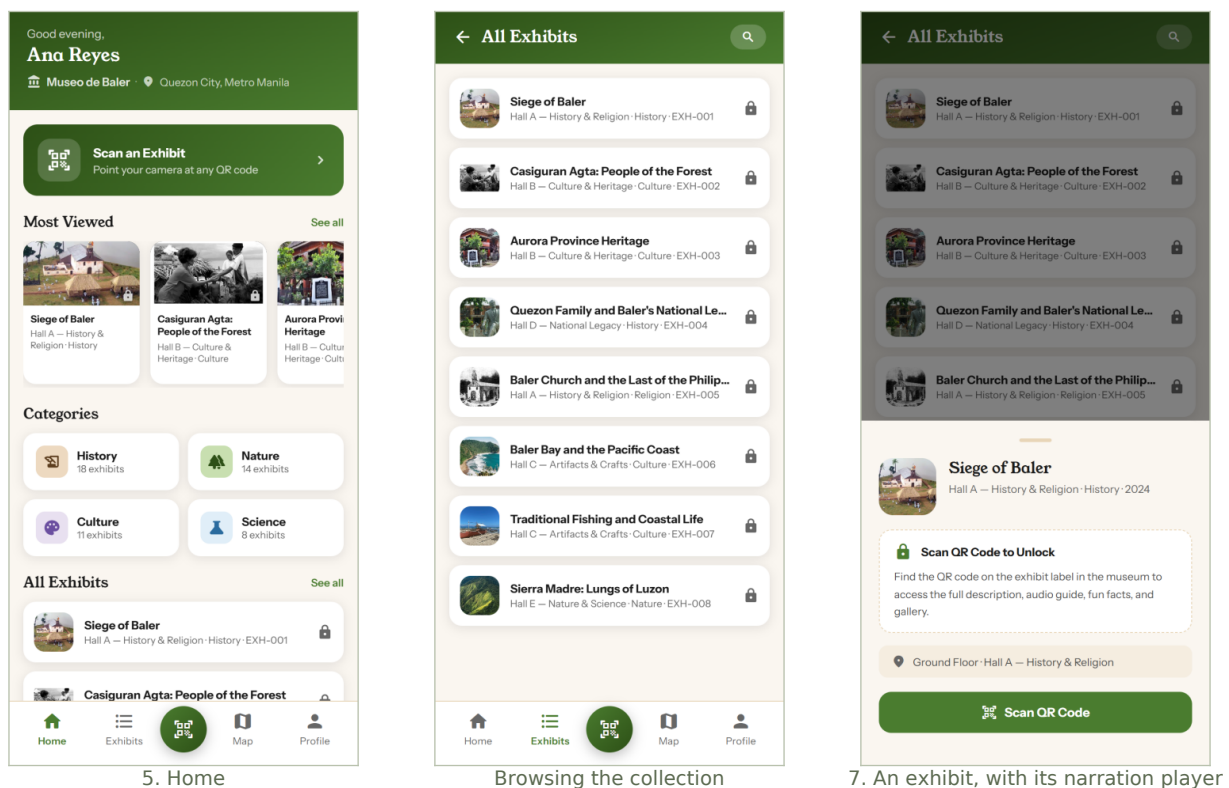
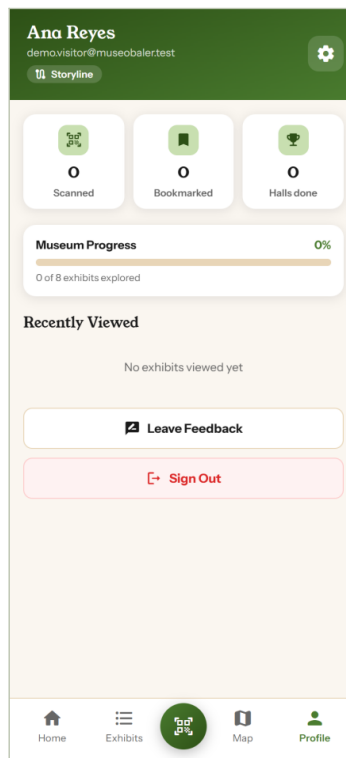


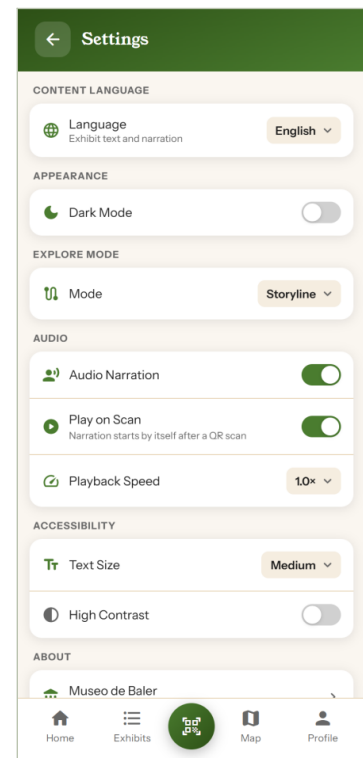
Figure A3.2 — Once cleared: the museum as the visitor moves through it.



8. The map, both floors



9. Progress: scanned, saved, halls covered



Settings (see A3.2)

Figure A3.3 — Finding the way, and what the visitor controls.

A3.2 Settings the visitor controls

Setting	Effect
Content language	Which language the exhibit text and narration are served in: English, Filipino or Spanish
Explore mode	Switches between Storyline and Free at any time
Audio narration, play on scan, playback speed	Whether narration starts by itself, and how fast it reads
Background music	Ambient audio, off by default
Text size, dark mode, high contrast	Accessibility controls; text size scales the whole interface, not only body text

A3.3 What the visitor keeps when the signal drops

Inside thick museum walls a phone often loses its connection. The app keeps its own copy of the screens, pictures, audio and the exhibits the visitor has already opened, and serves that copy when the network fails outright, so a tour does not end with an empty screen.

What deliberately does **not** work without a connection: registering, signing in, recording a scan, submitting feedback, and checking in at the geofence. Each says so plainly. There is no queue that replays these later — a scan recorded an hour late would put visitors in the wrong hall on the evening's report.

The app needs a secure connection. The camera, the location check and the offline copy all require the address to begin with <https://>. Opened over a plain local network address the app still runs, but with typed exhibit codes only and no attendance.

A4 · The front desk

Front Desk in the left menu, on the counter tablet or computer. This screen replaces the paper logbook, prices admission, and is the only place a visitor becomes cleared to see the collection.

A4.1 Three ways a visitor gets in

1. **On their own phone.** They scan the entrance poster, create an account, and the app tells them to come to the desk for their fee or ID check. Staff type nothing.
2. **On the counter tablet.** The same app on the museum's own tablet, turned round for the visitor to fill in.
3. **Staff type it.** For a visitor with no phone: **Register visitor** on the Front Desk page — name, visitor type, where they are from, and for a Baler resident, their barangay.

Front Desk

Generate Report Entrance poster Guided Tours

16 PEOPLE TODAY 4 ENTRIES LOGGED

One visitor
Someone arriving on their own, or without a phone

FIRST NAME * LAST NAME *

VISITOR TYPE *

Local Free check ID Tourist PHP 50.00 Foreign PHP 50.00

FROM (CITY OR COUNTRY) AGE

Baler, Quezon City, Japan...

Register visitor

A group arriving together
Four or five from out of town, a family, or a school tour — one entry, one payment

WHO IS SIGNING IN * HOW MANY *

Name of the person at the desk 4

GROUP TYPE * VISITOR TYPE *

Group of friends / co-travellers Tourist (other province)

HOW MANY ARE FROM BALER? (free — check their IDs) FROM

0 City or country

Collect PHP 200.00 — 4 paying of 4.

Register group

Staff check-in code
Staff scan this on arrival with their own phone

1. Open the admin panel on your phone and sign in
2. Tap **My Attendance**
3. Point your camera at this code

30s

Rotates every 60s location checked Open full screen →

Registered today

TIME	GROUP	CODE	JOINED	FROM BALER	FEE	PAYMENT
3:07 PM	Ana Aquino's group Group: 6 people Talar		0 / 6	AS 6	—	Free
10:56 AM	Ana Bautista's group Group: 6 people San Jose		0 / 6	0 of 6 Correct	PHP 300.00	Paid

TIME NAME TYPE HOW PAYMENT

10:49 PM	Miguel Santos	Tourist	App	Unpaid
10:49 PM	Ana Reyes	Tourist	App	Paid
10:56 AM	Jose Santos	Tourist	Desk	Unpaid
10:39 AM	Nestor Bautista	Local	App	Free

Administrator Museum Staff

Password Alerts Log out

Figure A4.1 — The Front Desk screen: registration, today's arrivals, and the group register with their payment state.

A4.2 Clearing a visitor

Nobody sees exhibit content until the desk clears them. Until then their screen shows the waiting message, and it unlocks the moment staff act.

Visitor type	What staff do	Result
Domestic tourist or foreign visitor	Collect the fee, then Mark Paid	Admission recorded at the current fee, with the staff member's name and the time. Their phone unlocks by itself.
Local resident of Baler	Sight a valid ID showing a Baler address, then Verify ID	Free admission, recorded with who verified it. If the ID is from elsewhere, they register as a tourist instead.

Both actions are available on the Front Desk page's *Registered today* list and in **Records**.

Museo de Baler
ADMIN PANEL

Records Generate Report

Visitor Records Groups Scan Records Attendance

47 TOTAL VISITORS 12 LOCAL 31 TOURIST

4 FOREIGN 8 FEES UNPAID 6 IDS TO VERIFY

1 visitor waiting to be let in
They cannot open the museum app until the desk collects the fee or checks their ID. Show queue

Search by name or country... All Visitor Types All Visit Types All Admissions Sort: Newest First

Today
Tuesday, 29 September 2026 · 4 visitors

NAME	AGE	SEX	VISIT TYPE	VISITOR TYPE	LAST VISIT	LOCATION	ADMISSION
Miguel Santos	29	Female	Walk-in	Tourist	Sep 29, 2026	Quezon City, Metro Manila	Waiting for payment Unpaid - P50.00 Mark Paid
Ana Reyes	29	Female	Walk-in	Tourist	Sep 29, 2026	Quezon City, Metro Manila	Unlocked Paid - P50.00
Jose Santos	16	—	Walk-in	Tourist	—	Baler	Waiting for payment Unpaid - P50.00 Mark Paid
Nestor Bautista	50	—	Walk-in	Local	—	San Jose	Unlocked Free entry: ID checked

Administrator
Museum Staff

Password Alerts Log out

Figure A4.2 — Records: everyone registered, with their payment state and the same Mark Paid and Verify ID actions.

A4.3 Groups: schools, families, tour agencies

A party is one record and one payment, not thirty registrations.

1. **Register group:** who is signing in, the group type, how many people, where they are from, and **how many of them are Baler residents** — those are free, so check their IDs.
2. The screen shows the total fee and a **six-letter code**. Read the code to the group leader. Anyone in the party who wants the app on their own phone enters that code when they register, and is counted inside the group instead of owing a second fee.
3. Collect the fee and **Mark Paid**. Every member's phone unlocks at once.

A group code cannot be abused. It works only on the day it was issued, and only for as many people as were counted. A code passed around outside cannot admit a sixth person to a party of five. Codes contain no letter O or I, so they cannot be misread as zero or one.

A4.4 Refunds and corrections

Situation	What to do
A Baler resident was charged inside a group, discovered after registering	On the group's row, change the <i>from Baler</i> count and click Correct . The group is re-priced. If it had already paid, the difference is recorded as a refund with the staff member's name — hand the cash back and the books match. Same-day only.
An individual visitor paid but should not have	There is no undo for an individual payment in the panel, on purpose. Note it in the logbook remarks and tell the Tourism office the same day; it is corrected in the records.
Marked paid by mistake, no money taken	The same: tell the Tourism office the same day.

A4.5 End of day

Records lists everyone from today with their payment state. **Records** → **Logbook** prints the day, or exports it as a spreadsheet or PDF, for the cash count (section A13).

A5 · Exhibits: add, edit, archive

Exhibits in the left menu. Every item a visitor can scan or find on the map is an exhibit.

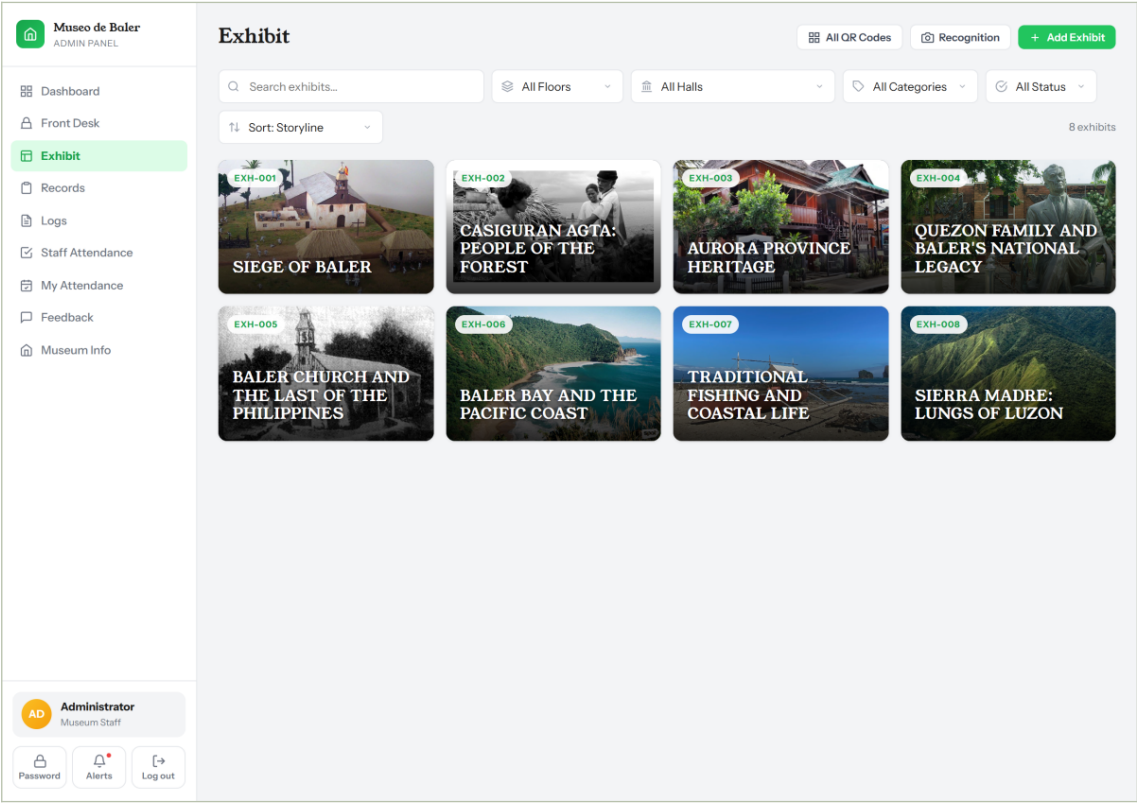


Figure A5.1 — The collection. The filter switches between active and archived pieces.

A5.1 Adding an exhibit

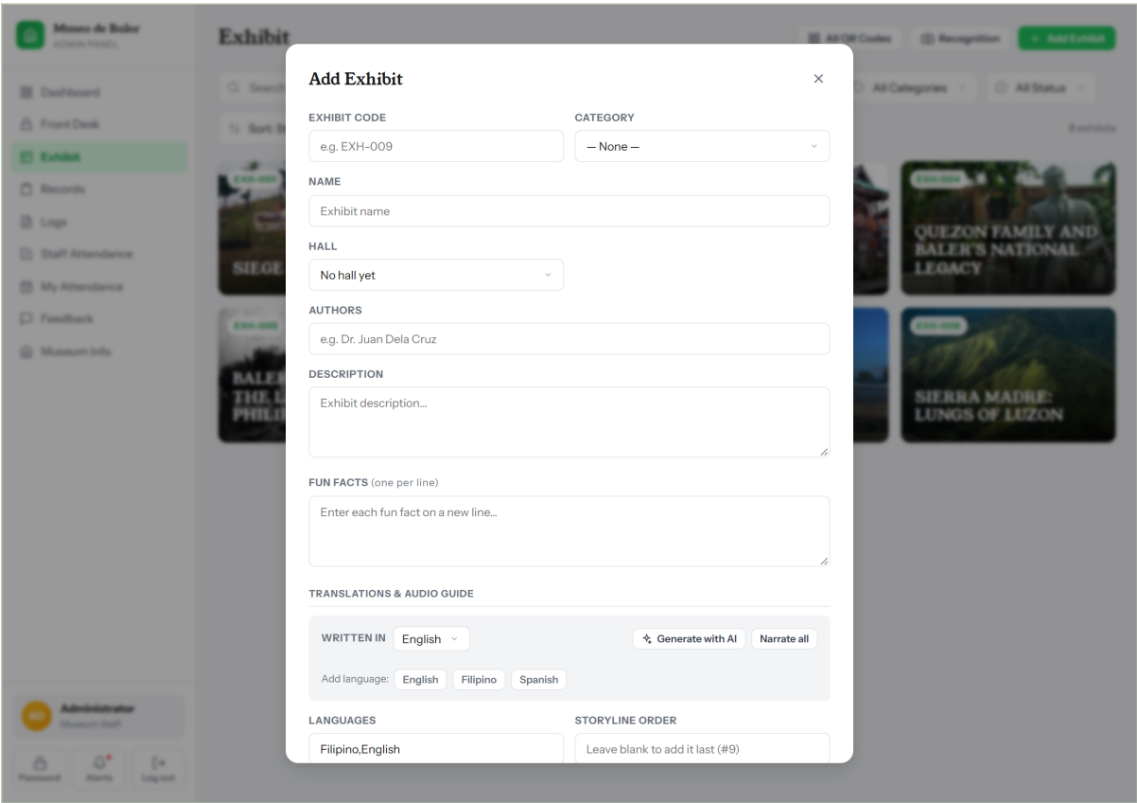


Figure A5.2 — The Add Exhibit form. Translations and narration are drafted on the same page and saved together with it.

Field	What it is for
Exhibit Code	Printed on the QR label and typed by visitors when the camera fails. Short, no spaces, unique, for example EXH-012. It cannot be changed after labels are printed without reprinting them.
Name and Description	What the visitor reads. Write these first; the translations and narration are made from them.
Fun Facts	One per line. Leave it empty rather than padding it — the app hides the tab when there are none.
Category, Hall	The hall list comes from Museum Info (section A9).
Authors	Who curated or wrote it; shown to visitors.
Languages	Which languages this exhibit is offered in.
Storyline Order	The number of this stop on the guided path (1, 2, 3 ...). Leave blank for an exhibit that is not on the path.
Image	JPG, PNG, GIF or WebP up to 10 MB. This is the picture visitors see and the one the camera compares against.

If the form comes back with a red message, fix the field it names and **choose the picture again** — browsers do not keep a selected file when a form is sent back.

A5.2 Editing an exhibit

Museo de Baler
ADMIN PANEL

Edit Exhibit
Siege of Baler

[← Back to Exhibits](#)

Exhibit Details

EXHIBIT CODE: EXH-001 CATEGORY: History

NAME: Siege of Baler

HALL: Hall A — History & Religion · G

AUTHORS: Museo de Baler Curatorial Team

DESCRIPTION:

From July 1, 1898 to June 2, 1899, a garrison of 54 Spanish soldiers under Captain Enrique de las Morenas fortified themselves inside the San Luis Obispo de Tolosa Church in Baler. Filipino revolutionary forces laid siege for 337 days — one of the longest last stands of the Spanish colonial era. Remarkably, the soldiers held out even after Spain had already ceded the Philippines to the United States through the Treaty of Paris on December 10, 1898. The siege ended when Lieutenant Saturnino Martin Cerezo finally surrendered, unaware the war had long been over. This exhibit features a diorama of the siege, period weapons, and historical documents.

FUN FACTS (ONE PER LINE)

The siege lasted 337 days — the longest holdout of the Spanish colonial period in the Philippines.
The Spanish soldiers were unaware that Spain had already surrendered the Philippines to the US months earlier.
Captain Enrique de las Morenas died of illness during the siege; Lieutenant Martin Cerezo took command.
Only 33 of the original 54 soldiers survived to surrender on June 2, 1899. The event is commemorated annually on Philippine-Spanish Friendship Day.

LANGUAGES: Filipino, English STORYLINE ORDER: 1

IMAGE:

[Choose File](#) No file chosen

[Cancel](#) [Save Changes](#)

Translations & Audio [+ Add](#)

English (en) [+ Add](#)

Siege of Baler

▶ 0:00 / 0:39 [Speaker](#) [More](#)

Filipino (fi) [+ Add](#)

Pagkubkob ng Baler

▶ 0:00 / 0:52 [Speaker](#) [More](#)

Camera Recognition

0 training photo(s) [needs 30+ to be recognised well](#) [Manage photos](#)

Gallery Images

ADD IMAGES

[Choose Files](#) No file chosen

CAPTION (OPTIONAL)

e.g. Front view

[Upload](#)

Administrator
Museum Staff

[Password](#) [Alerts](#) [Log out](#)

Figure A5.3 — The edit page also carries Translations & Audio, Camera Recognition photos, and the gallery.

Open the exhibit, click **Edit**, change what you need, and **Save Changes**. The same page holds **Translations & Audio** (section A6), **Camera Recognition** photos (section A10), and **Gallery Images** — extra pictures with captions shown in the app's gallery tab. Changes reach visitors' phones the next time the app refreshes its list, usually within a minute.

A5.3 Archive and restore

Archiving hides an exhibit from visitors and from the map without deleting anything — its scans, photos and translations stay. Open the exhibit and click **Archive**; it disappears from the app at once and from the QR sheet. To bring it back, set the filter on the Exhibits page to **Archived**, open it, and click **Restore**.

Use archive for a piece on loan or in conservation. **There is no button that deletes an exhibit outright, on purpose** — its scan history and printed QR labels outlive it.

A6 · Translations and the audio guide

Each exhibit can carry a title, description and spoken narration in several languages; visitors pick their language in the app. The panel supports **English**, **Filipino** and **Spanish**. There are two ways to fill these in, and they can be mixed.

A6.1 Method A: let the AI draft them

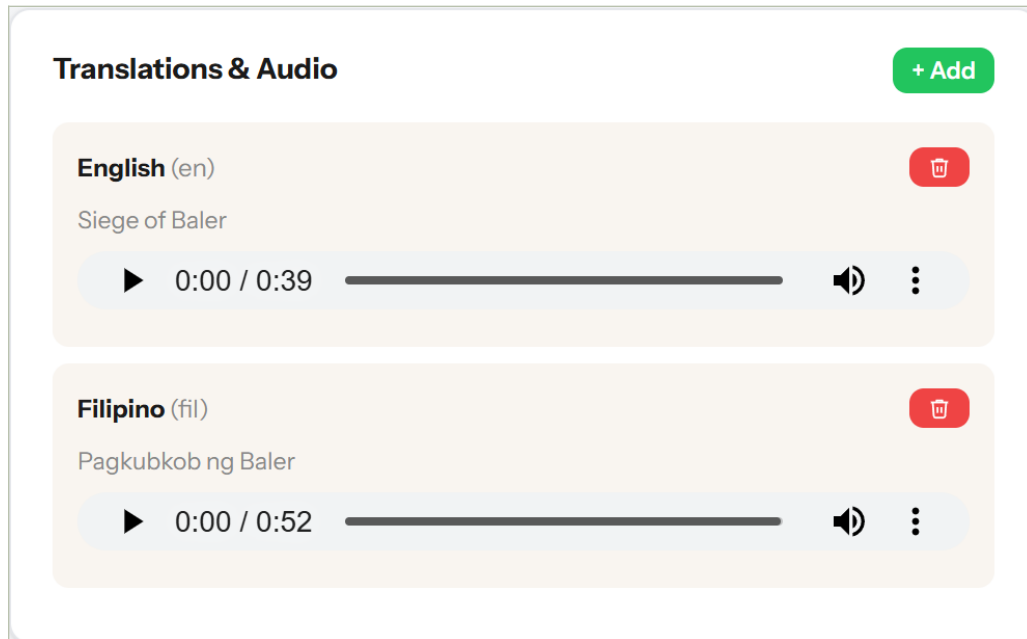


Figure A6.1 — The Translations & Audio Guide section. Drafts are marked *AI draft — review* until a curator has read them.

1. Write the **Name**, **Description** and **Fun Facts** first — that is what gets translated.
2. Click **Translate**. A card appears per language, marked *AI draft — review*. **Read each one**. The AI is good but it is not a curator: check names, dates and anything with a local meaning. Edit the text in the card directly.
3. Click **Narrate all**, or **Narrate** on one card. This takes a few seconds per language. A player appears — **listen before saving**.
4. If you edit a card's text after narrating it, the card warns *Text changed since this was narrated*. Narrate again, or the voice will read the old text.
5. **Nothing is stored until you save the exhibit**. Closing the form discards the drafts.

Limits, so nobody is surprised

- **There is a budget:** 20 requests a minute and 300 a day per staff account. Past that the button says to wait; the daily count resets the next day. Narrating three languages for one exhibit is three requests.
- **It needs the internet and a Google AI key** configured on the server. If the button says *AI generation is not set up*, that key is missing.
- **If it says the text was declined**, the wording tripped Google's content filter. Reword and try again; the draft is not saved.
- **The museum works without it**. With no key at all, every other feature is unaffected and translations are typed by hand.

A6.2 Method B: type a translation and upload a recording

On the edit page, **Translations & Audio** → **+ Add**:

1. **Code** (en, **fil**, es) and **Label** — the name visitors see.
2. **Title** and **Description** in that language.
3. **Audio** (MP3 or WAV) — a recording by a staff member or a hired narrator. Keep it to a few minutes; the app streams it, and a visitor on weak Wi-Fi waits for the whole file.
4. **Save**. To replace a recording, delete the translation and add it again.

The two methods mix freely: AI text with a human recording, or a human translation with AI narration.

A7 · QR codes and the entrance poster

Every active exhibit has a QR code that opens it in the app. Visitors scan it with the app's scanner, or with their phone's own camera, which opens the app.

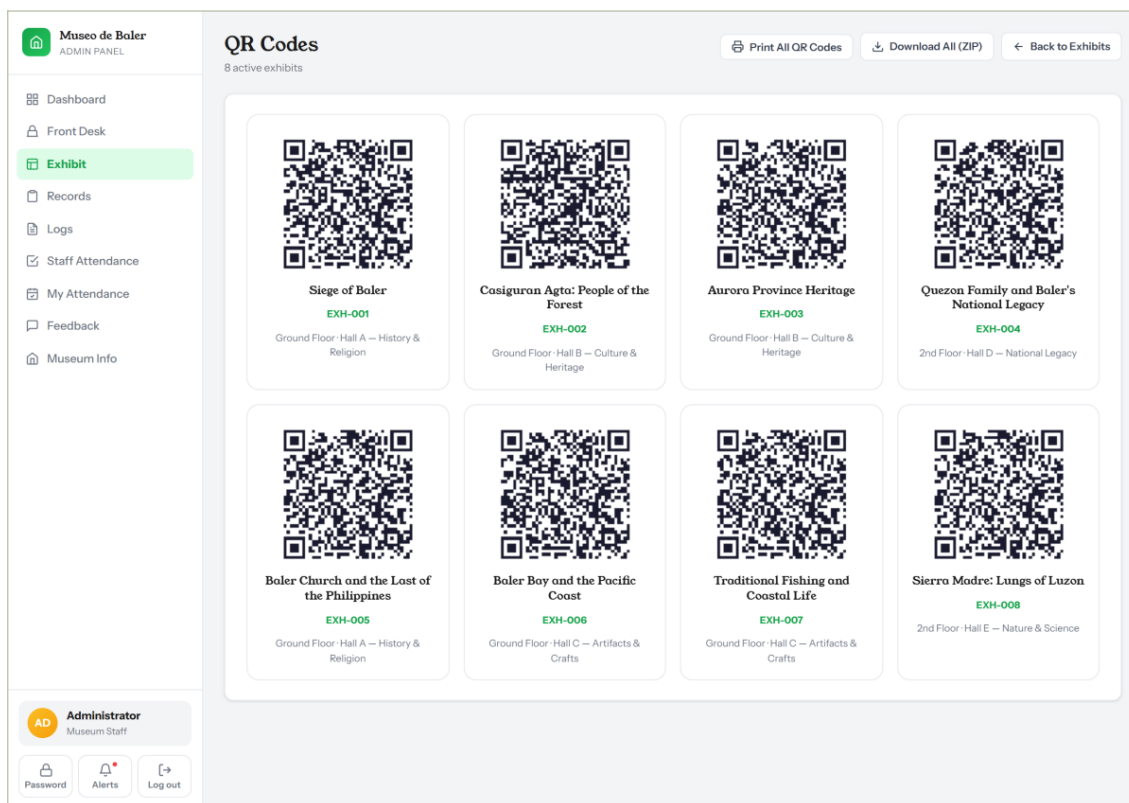


Figure A7.1 — The QR Codes page: one label per active exhibit, printable as a sheet or downloadable individually.

Option	What you get
Download SVG (one exhibit)	A single label as a vector file. Print it at any size; it stays sharp.
Print All QR Codes	A sheet of every active exhibit, from the browser's print dialog. Plain paper or sticker sheets.
Download All (ZIP)	Every label as a separate vector file in one download, for a print shop.

Each label carries the exhibit's code in plain text under the QR square, so a visitor whose camera will not focus can type it instead.

Where to put them

At reading height, in the light, and not behind glass that reflects. A label under a spotlight glare will not scan; the app then tells the visitor to type the code.

The entrance poster

Front Desk → Poster is a different QR code: it opens the app's sign-in page, for visitors who arrive without having installed anything. Print it and stand it at the door. This is how self-registration happens when there is no counter tablet.

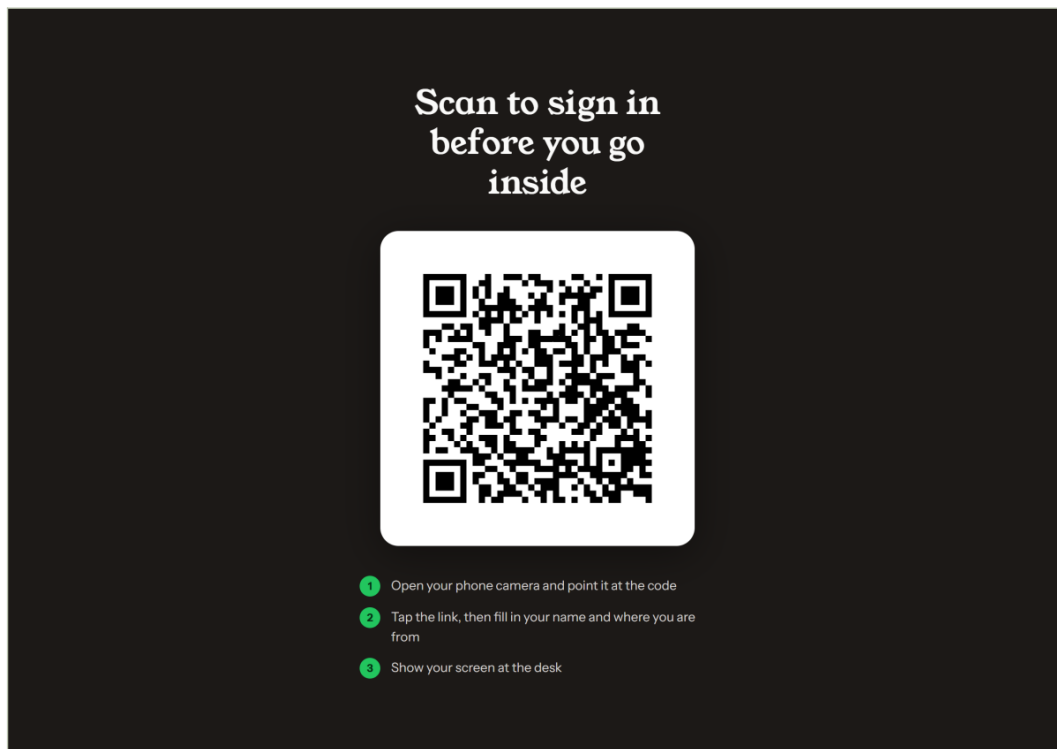


Figure A7.2 — The printable entrance poster. This is the first thing most visitors ever touch.

A8 · The museum map and the storyline

Map in the menu. Visitors see the same map in the app, with the exhibits they have found marked and, in storyline mode, the path from stop to stop.

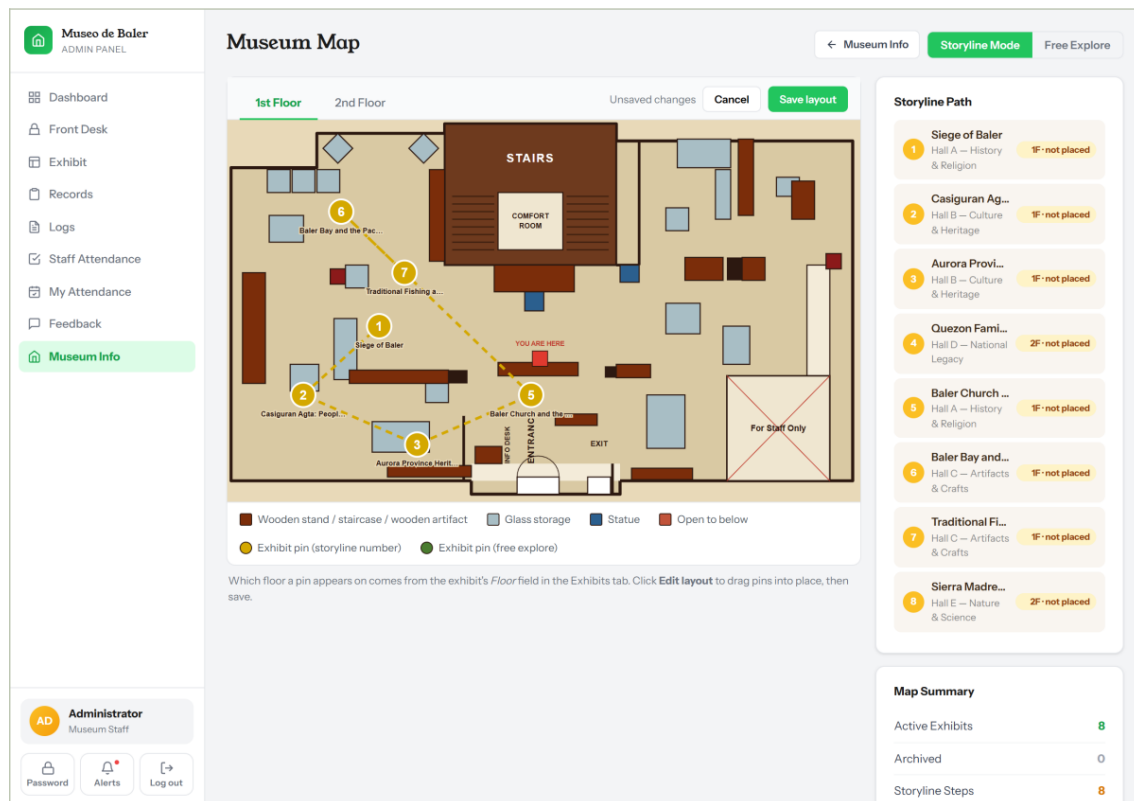


Figure A8.1 — The map editor. Pins are dragged onto the floor plan and saved; the storyline path is listed alongside.

1. Switch between **1st Floor** and **2nd Floor** at the top.
2. Click **Edit layout**.
3. **Drag** each exhibit pin to where the piece actually stands. An exhibit that has never been placed appears in a default spot; put it right.
4. Click **Save layout**. Nothing moves for visitors until you do.
5. The **Storyline Path** panel lists the stops in order and warns when an exhibit is missing a storyline number — set that in the exhibit's own form (section A5).

The floor plan drawing itself — the walls, rooms and stairs — is part of the software, not something staff redraw here. Exhibit *pins* are data and staff own them. If a wall moves, the maintainer edits the drawing; see the technical documentation.

A9 · Museum info: hours, fee, halls, the geofence

Museum Info in the menu. This is what visitors read on the app's About screen and the numbers the desk works from. Click **Edit**, change what you need, then **Done**.

Museum Info

Museo de Baler
Baler, Aurora, Philippines

Basic Information
Name, tagline, and story

MUSEUM NAME: Museo de Baler

TAGLINE / LOCATION: Baler, Aurora, Philippines

OUR STORY (FIRST PARAGRAPH)
Museo de Baler is the official municipal museum of Baler, Aurora. Inaugurated in 2002, it was established to commemorate Philippine-Spanish Friendship Day and the historic Siege of Baler. The museum preserves and showcases the rich cultural heritage, natural history, and significant historical events of Baler and Aurora Province.

SECOND PARAGRAPH
The museum houses artifacts, dioramas, and exhibits spanning the pre-colonial era, the Spanish colonial period, the Philippine Revolution, and the modern history of Aurora Province — including the birthplace of President Manuel L. Quezon, the first President of the Philippine Commonwealth.

Contact & Hours
Address, schedule, and contact details

Report Branding
The letterhead printed at the top of every report

Geofencing
GPS anchor point the visitor app uses for auto check-in

Museum Halls
5 halls configured

Administrator
Museum Staff

Password Alerts Log out

Figure A9.1 — Museum Info: the About screen text, the admission fee, the halls, and the geofence that both the visitor check-in and staff clock-in read.

A9.1 The fields

Field	Notes
Museum Name, Tagline, Our Story, Second Paragraph	The app's About screen.
Address, Opening Hours, Closed On, Phone, Email	Also the About screen. Keep <i>Closed On</i> current — the app shows it and visitors plan by it.
Admission Fee (PHP, per visitor)	The flat fee for tourists and foreign visitors; Baler residents are free. Change it here and the desk, the app and the reports all use the new amount from the next registration onward. It does not change what people have already paid. The sentence <i>shown to visitors as</i> is written from the number, so it can never disagree with what the desk collects.
Museum Halls	The halls exhibits are assigned to. Add one with a name, floor and description. Delete one only when no exhibit uses it.

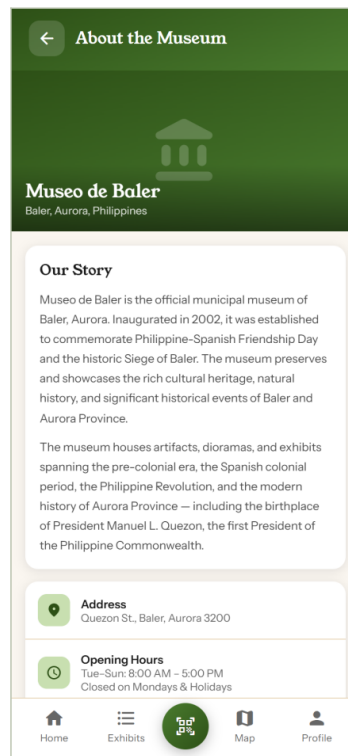


Figure A9.2 — The same information as visitors read it, on the app's About screen. Everything here comes from the fields above, so changing one changes the other.

A9.2 The geofence

The app records a visitor as being in the museum when their phone is inside a circle drawn around the building, and staff can only clock in from inside that same circle.

Field	Notes
Latitude, Longitude	The centre of the circle. Set it once, from a phone standing at the entrance: on the phone, open My Attendance → Use my current location .
Geofence Radius (metres)	150 m covers the building and its forecourt. Larger catches phones out on the street; smaller loses people at the back of the building.

Do not set the pin from a desktop computer. A desktop guesses its position from Wi-Fi, and during testing that put the pin 1.8 km from the door — after which nobody could clock in and no visitor was counted as present. This is why the panel offers a phone button for it.

A10 · Camera recognition

Besides the QR labels, the app can identify an exhibit through the phone camera. It works only after staff have taught it, and it fails softly: with no training the app falls back to a weaker server-side match, and **the QR codes always work regardless**.

Museo de Baler ADMIN PANEL

Exhibit Recognition [← Back to Exhibits](#)

Recognition is on
Last trained Sep 22, 2026 · 8:33 PM · knows 9 exhibit(s)

The model still knows codes that are no longer active exhibits (EXH-009). Training again will drop them.

8 exhibit(s) have fewer than 20 photos. They will be recognised poorly, or not at all, until more are taken.

Only 0 background photo(s). Without walls, floors and empty cases to learn from, the model has to guess an exhibit for every frame. Add background photos.

Train the model
Takes a minute or two on this computer. Visitors get the new model the next time they open the scanner.
[Train](#)
Ready.

EXHIBIT	CODE	TRAINING PHOTOS	IN CURRENT MODEL	
Siege of Baler Hall A — History & Religion	EXH-001	0 too few	yes	Manage photos
Casiguran Agta: People of the Forest Hall B — Culture & Heritage	EXH-002	0 too few	yes	Manage photos
Aurora Province Heritage Hall B — Culture & Heritage	EXH-003	0 too few	yes	Manage photos
Quezon Family and Baler's National Legacy Hall D — National Legacy	EXH-004	0 too few	yes	Manage photos
Baler Church and the Last of the Philippines Hall A — History & Religion	EXH-005	0 too few	yes	Manage photos
Baler Bay and the Pacific Coast Hall C — Artifacts & Crafts	EXH-006	0 too few	yes	Manage photos
Traditional Fishing and Coastal Life Hall C — Artifacts & Crafts	EXH-007	0 too few	yes	Manage photos
Sierra Madre: Lungs of Luzon	EXH-008	0 too few	yes	Manage photos

Administrator
Museum Staff

[Password](#) [Alerts](#) [Log out](#)

Figure A10.1 — The Recognition page, showing how many training photos each exhibit has and the button that trains the model.

1. **Recognition** in the menu. Each exhibit shows its photo count: it needs **at least 20, and 30 or more to work well**.
2. **Manage photos** on an exhibit. Do this **on a phone** — this page is deliberately allowed on a phone because that is where the camera is. Stand where a visitor would stand and take photos from several angles, distances and times of day.
3. **Background photos**: 20–30 pictures of walls, floors, corridors and other objects — things that are *not* an exhibit — so the camera learns to say “nothing here” instead of guessing.
4. Back on a computer, click **Train the model**. The training runs in the browser and takes a minute or two; **do not close the tab**. Every phone picks up the new model the next time the app refreshes.

Retrain after adding an exhibit, moving one to a different-looking spot, or changing its lighting.

A11 · Guided tours and visitor feedback

A11.1 Tours

Most visits are unguided, so this screen is deliberately small. **Start tour** — pick the guide and who the tour is for (one visitor or a group). **End tour** when it is done. A visitor who was guided is asked in the app's feedback sheet how their guide was; nobody else is shown that question.

Museo de Baler
ADMIN PANEL

Dashboard
Front Desk
Exhibit
Records
Logs
Staff Attendance
My Attendance
Feedback
Museum Info

Administrator
Museum Staff

Password Alerts Log out

Guided Tours

← Front Desk 09/01/2026 to 09/29/2026 Show

Assign a guide
Only staff who have checked in today can be assigned.

GUIDE: Ana Reyes WHY: Visitor asked FOR WHOM: Choose... PAX: auto

Start tour

Tours Sep 1 - Sep 29, 2026 6 total

DATE	GUIDE	TYPE	FOR	PAX	DURATION
Sep 29, 11:05 AM	Ana Reyes	Foreign	Ramon Bautista	22	41 min
Sep 28, 11:05 AM	Jun Molina	Foreign	Migs Lim	9	43 min
Sep 25, 12:47 PM	Marites Domingo	Requested	Liza Bautista	3	72 min
Sep 23, 2:34 PM	Marites Domingo	Educational	Paolo Aquino	3	59 min
Sep 22, 10:42 AM	Ana Reyes	Foreign	Liza Cruz	9	53 min
Sep 18, 1:41 PM	Jun Molina	Requested	Ramon Lim	6	44 min

Figure A11.1 — Guided tours: who is guiding, who is being guided, and when it ended.

A11.2 Feedback and the ARTA survey

Feedback lists what visitors said, together with the ARTA Client Satisfaction Measurement survey their answers feed. **Reports** → **Feedback** produces the CSM figures for a date range and the file the LGU filing needs (section A13). The Tourism office owns the questions themselves (section A12.4).

Museo de Baler
ADMIN PANEL

Dashboard
Front Desk
Exhibit
Records
Logs
Staff Attendance
My Attendance
Feedback
Museum Info

AD Administrator
Museum Staff

Password
Alerts
Log out

Feedback

Generate Report

Total Feedback: 28
Average Rating: 4.2 / 5
CSM Score: —
CC Awareness: —

Search by name or keyword...
All Ratings
Sort: Newest First

NAME	RATING	SURVEY	FEEDBACK	DATE	ACTION
Jose Santos	★★★★☆	—	Small but worth the visit.	Sep 29, 2026	View
Ramon Bautista	★★★★★	—	Our guide explained the history really well.	Sep 29, 2026	View
Nestor Bautista	★★★★★	—		Sep 29, 2026	View
Migs Lim	★★★★☆	—	Our guide explained the history really well.	Sep 28, 2026	View
Paolo Reyes	★★★★☆	—	Small but worth the visit.	Sep 27, 2026	View
Migs Bautista	★★★★★	—	The audio guide in Filipino was a nice touch.	Sep 26, 2026	View
Paolo Lim	★★★★☆	—		Sep 26, 2026	View
Jose Aquino	★★★★★	—	Very informative, enjoyed the Siege of Baler exhi...	Sep 26, 2026	View
Ramon Garcia	★★★★★	—	Very informative, enjoyed the Siege of Baler exhi...	Sep 26, 2026	View
Liza Bautista	★★★★☆	—	Our guide explained the history really well.	Sep 25, 2026	View
Jose Santos	★★★★★	—	Hard to find parking, but the museum itself was g...	Sep 25, 2026	View
Liza Cruz	★★★★☆	—	Hard to find parking, but the museum itself was g...	Sep 24, 2026	View
Nestor Cruz	★★★★☆	—		Sep 23, 2026	View
Bea Santos	★★★★★	—	Very informative, enjoyed the Siege of Baler exhi...	Sep 23, 2026	View
Paolo Aquino	★★★★★	—	Our guide explained the history really well.	Sep 23, 2026	View
Ramon Lim	★★★★☆	—		Sep 22, 2026	View
Cristina Santos	★★★☆☆	—	Clean and well kept. Wish there were more seatin...	Sep 22, 2026	View
Liza Cruz	★★★★☆	—	Very informative, enjoyed the Siege of Baler exhi...	Sep 22, 2026	View
Liza Cruz	★★★★☆	—	Our guide explained the history really well.	Sep 22, 2026	View
Migs Lim	★★★★☆	—	Small but worth the visit.	Sep 20, 2026	View

28 responses

Previous
1
2
Next

Figure A11.2 — Visitor feedback, with the ARTA Client Satisfaction figures the answers roll up into.

A12 · The Tourism office's screens

Staff accounts, schedules, correction approvals and the survey question bank belong to the Tourism Head role and are not visible to museum staff at all. The audit trail at the end is the exception: both roles can read it, and only Tourism can export it.

A12.1 Staff accounts

Staff → Add Staff: full name, email address, role. **The system generates the password and shows it once** — copy it and hand it over in person. The staff member must change it at first sign-in.

- **Reset password** on an existing account does the same for someone who has forgotten theirs. It is a separate action from editing the account, so it gets its own audit entry and cannot happen by accident while someone is correcting a spelling.
- **Toggle** disables an account without deleting it — for someone who has left. Their past actions stay in the records.

NAME	EMAIL	ROLE	STATUS	PASSWORD	CREATED	ACTIONS
Ana Reyes	ana@museobaler.com	Museum Staff	Active	Temporary	Sep 29, 2026	Edit, Reset password, Deactivate
Marites Domingo	marites@museobaler.com	Museum Staff	Active	Temporary	Sep 29, 2026	Edit, Reset password, Deactivate
Jun Molina	jun@museobaler.com	Museum Staff	Active	Temporary	Sep 29, 2026	Edit, Reset password, Deactivate
Rosa Villanueva	rosa@museobaler.com	Museum Staff	Active	Temporary	Sep 29, 2026	Edit, Reset password, Deactivate
Administrator	admin@museobaler.com	Museum Staff	Active	Their own	Sep 29, 2026	Edit, Reset password, Deactivate
Tourism Head	tourism@baler.gov.ph	Tourism Head	Active	Their own	Sep 29, 2026	Edit

Figure A12.1 — Staff accounts. Museum staff cannot open this screen at all.

A12.2 Staff attendance

Museum staff clock in by scanning the code on the staff-room tablet (**Attendance → Kiosk**, left running) with their own phone, from inside the geofence. **The kiosk code rotates every 60 seconds**, so a photograph of it is worthless.

- **Staff Attendance** shows the sheet for a period.
- **DTR** prints or exports the month for one person (section A13).
- **Schedule**, per staff member, is what decides who counts as late.

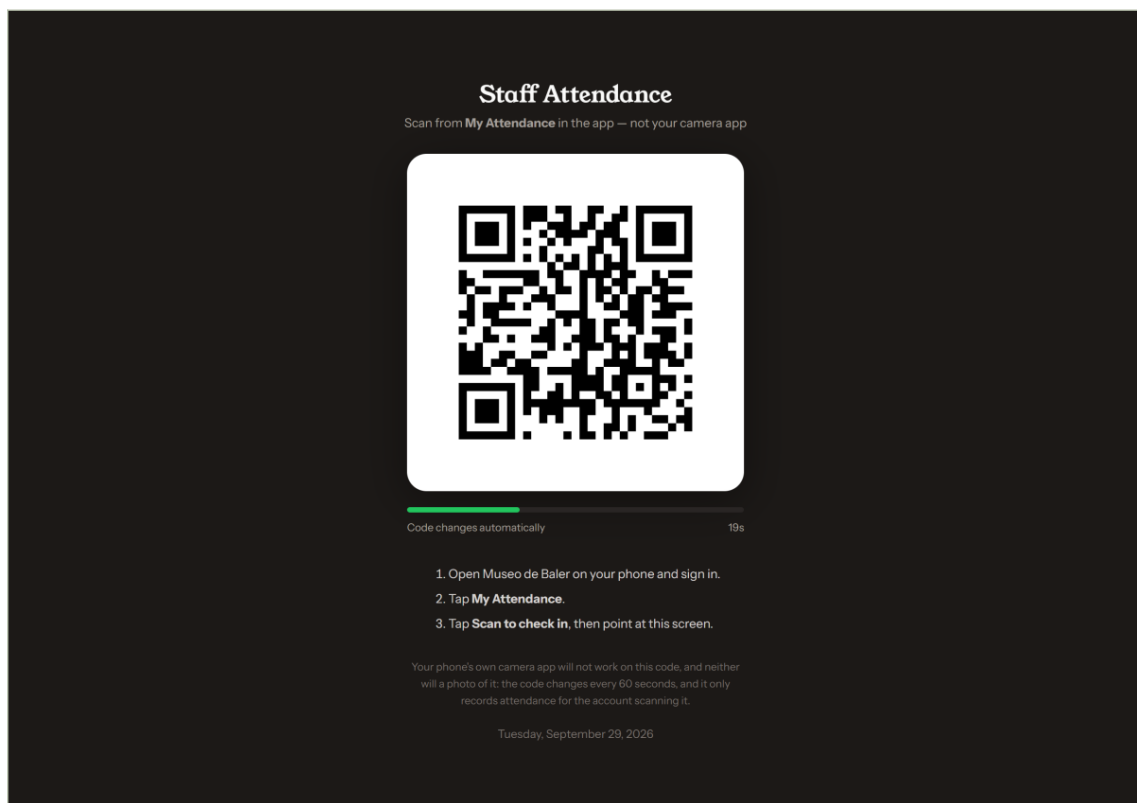


Figure A12.2 — The staff-room kiosk. The code changes every 60 seconds, so a photograph of it is worthless.

STAFF	ROLE	SHIFT	IN	OUT	WORKED	STATUS	Month
Administrator	Museum Staff	8:00 AM-5:00 PM	—	—	—	Absent	Month
Ana Reyes	Museum Staff	8:00 AM-5:00 PM	7:41 AM	—	—	Present	Month
Jun Molina	Museum Staff	8:00 AM-5:00 PM	7:47 AM	—	—	Present	Month
Marites Domingo	Museum Staff	8:00 AM-5:00 PM	7:49 AM	—	—	Present	Month
Rosa Villanueva	Museum Staff	8:00 AM-5:00 PM	7:45 AM	—	—	Present	Month

Figure A12.3 — The attendance sheet, and the route to one person's DTR.

A12.3 Attendance corrections

If someone forgot to clock in, **museum staff file a correction** with the reason, and **the Tourism office approves or rejects it**. Nobody can do both halves, and nobody can file one for themselves.

Both roles can read the list; only Tourism can act on it.

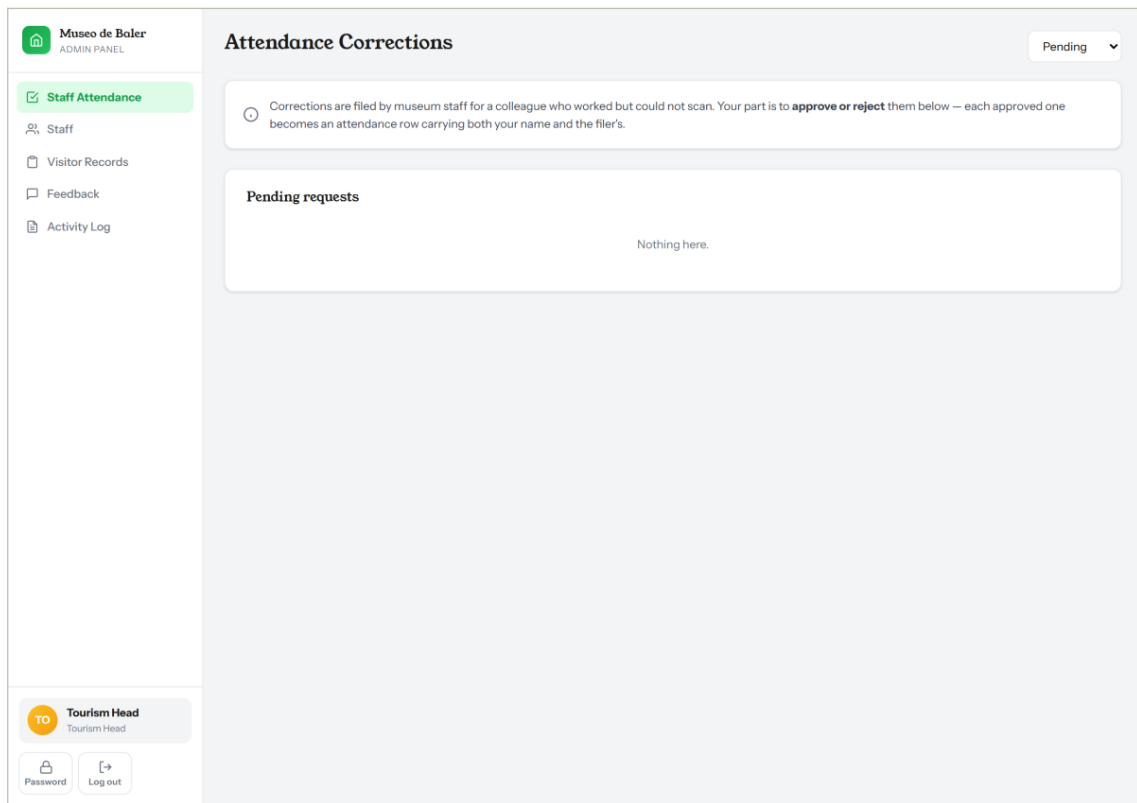


Figure A12.4 — Corrections waiting on the office. The person who filed one can never be the person who approves it.

A12.4 The visitor survey

Survey edits the question bank the app shows visitors. The questions are the ARTA Client Satisfaction Measurement form the office files with the LGU, so the office that answers for the numbers owns the questions. The standard ARTA set is locked; extra questions can be added or retired. **A retired question keeps its answers in past reports** — retiring it changes what new visitors are asked, never what old reports say.

Museo de Baler
ADMIN PANEL

- Staff Attendance
- Staff
- Visitor Records
- Feedback**
- Activity Log

Tourism Head
Tourism Head

Password Log out

Survey Questions

← Feedback Restore defaults Save changes

The first twelve questions are the **ARTA CSM form** (CC1-CC3, SQD0-SQD8). Their wording can be edited and any question can be reordered by dragging, but the ARTA questions keep their code and cannot be removed or switched off — the report is only a CSM report while every one of them is in it. Everything in **About the museum & app** is yours: add, edit, switch off or delete. The app picks up changes the next time a visitor opens the survey.

Citizen's Charter

Step 1 in the app: CC1-CC3: single choice

+ Add question

- CC1** Which of the following best describes your awareness of the Citizen's Charter (CC)? CHOICE ARTA
- CC2** Would you say that the CC of this office was... CHOICE ARTA
- CC3** How much did the CC help you in your transaction? CHOICE ARTA

Service Quality Dimensions

Step 2 in the app: SQD0-SQD8: five faces + N/A

+ Add question

- SQD0** I am satisfied with the service that I availed. FACES ARTA
- SQD1** I spent a reasonable amount of time for my transaction. FACES ARTA
- SQD2** The office followed the transaction's requirements and steps based on the information provided. FACES ARTA
- SQD3** The steps (including payment) I needed to do for my transaction were easy and simple. FACES ARTA
- SQD4** I easily found information about my transaction from the office or its website. FACES ARTA
- SQD5** I paid a reasonable amount of fees for my transaction. FACES ARTA
- SQD6** I feel the office was fair to everyone, or "walang palakasan", during my transaction. FACES ARTA
- SQD7** I was treated courteously by the staff, and (if asked for help) the staff was helpful. FACES ARTA
- SQD8** I got what I needed from the government office, or (if denied) the denial was sufficiently explained to me. FACES ARTA

About the museum & app

Step 3 in the app: shown with the star rating and comment

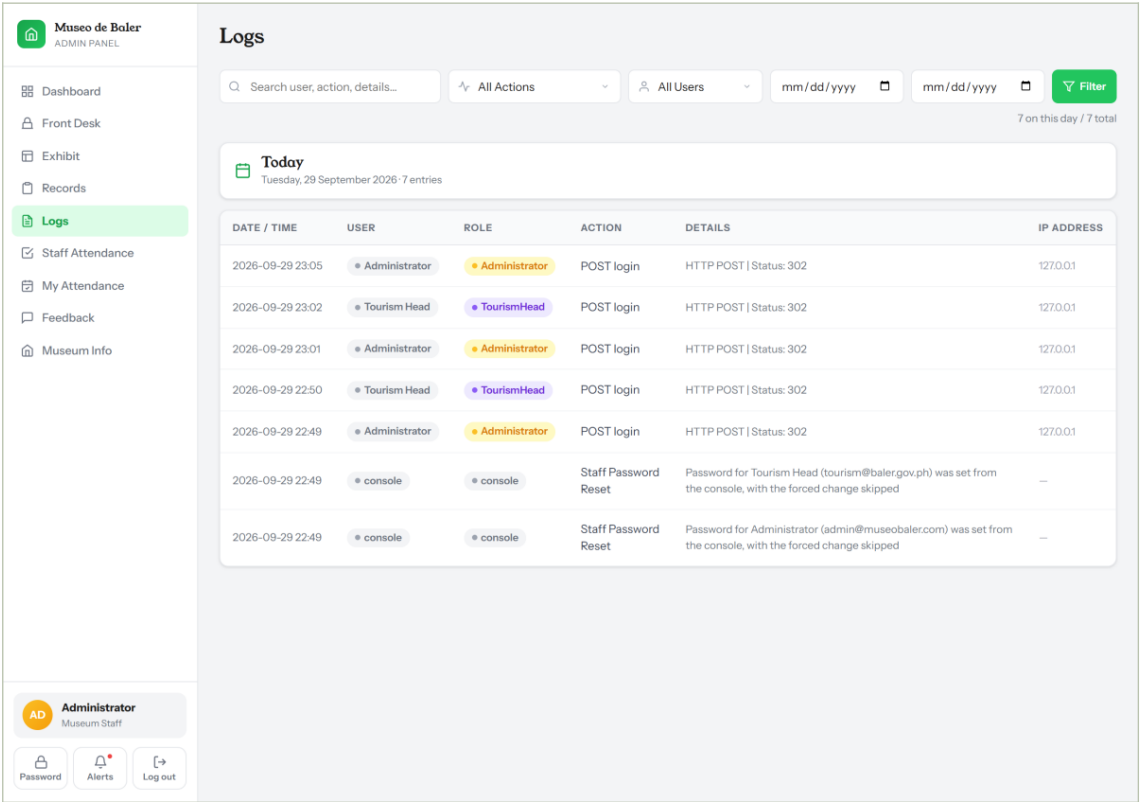
+ Add question

- APP_EASE** The Museo de Baler app was easy to use. FACES
- APP_CONTENT** The exhibit pages and audio guide in the app helped me understand the exhibits. FACES
- APP_CHECKIN** QR check-in with the app was smooth and quick. FACES

Figure A12.5 — The ARTA question bank. The standard set is locked.

A12.5 The audit trail

Every change anyone makes in the panel is recorded: who, what, from where and when. Both roles can read it; **only the Tourism office can export it**, because it is the record kept on the museum staff as well as by them.



A13 · Reports and exports

There is no single reports hub page. Each report is reached from the section it belongs to — the logbook from Records, the DTR from Staff Attendance — because that is where the person who needs it already is.

A13.1 The six reports and their formats

Every report can be previewed on screen before committing to a download, so nobody spends a download working out whether the date range was right.

The screenshot displays the 'Daily Visitor Logbook' interface for Tuesday, September 29, 2026. At the top, there are buttons for 'Back', 'Save', 'Print', 'SAVE AS', and 'PDF'. Below this is a 'Preview — PDF' section showing a thumbnail of the report. The main report area is titled 'Museo de Baler Daily Visitor Logbook' and includes the following data:

TOTAL PEOPLE 16
ENTRIES 6
COLLECTED PHP 350.00
OUTSTANDING PHP 100.00

Groups

TIME	GROUP / CONTACT	TYPE	PAX	FROM	FEE	PAYMENT
10:56 AM	Ana Bautista	Tourist	6	San Jose	300.00	Paid
3:07 PM	Ana Aquino	Local	6	Tarlac	—	Free
Group subtotal					12	300.00

Individual visitors

TIME	NAME	TYPE	AGE	FROM	ENTERED	FEE	PAYMENT
10:39 AM	Nestor Bautista	Local	50	San Jose	App	—	Free
10:56 AM	Jose Santos	Tourist	16	Baler	Desk	50.00	Unpaid
10:49 PM	Ana Reyes	Tourist	29	Quezon City	App	50.00	Paid
10:49 PM	Miguel Santos	Tourist	29	Quezon City	App	50.00	Unpaid

Generated September 29, 2026 11:06 PM by Administrator. Figures are taken from the visitor records for this date. "Entered" shows whether the visitor registered on their own phone, on the counter tablet, or was keyed in by staff.

At the bottom, there are fields for 'Prepared by' (Front desk), 'Verified by' (Museum Administrator), and 'Noted by' (Municipal Tourism Office).

Figure A13.1 — The daily logbook, with the toolbar that previews and downloads it in each format the report allows.

Report	What it contains	CSV	XLSX	DOCX	PDF
Logbook	One day's arrivals, individuals and groups, with payment state — the cash count	Yes	Yes	—	Yes

Report	What it contains	CSV	XLSX	DOCX	PDF
DTR	One staff member's daily time record for a month	Yes	Yes	—	Yes
Visitors	Visitor counts and composition over a date range	Yes	Yes	—	Yes
Exhibits	The collection and how often each piece was scanned	Yes	—	Yes	Yes
Feedback	Visitor feedback and the ARTA CSM figures	Yes	—	Yes	Yes
Audit trail <i>(Tourism only)</i>	Every change made in the panel, by whom and when	Yes	—	—	Yes

Why the format list differs per report. The formats offered are the ones each report is actually wanted in. A DTR and a logbook are columns of figures that an office re-totals, so they offer a spreadsheet. The exhibit and feedback reports are narrative documents that get pasted into a larger submission, so they offer Word. Every report offers CSV for machines and PDF for filing. An unsupported combination is refused by the system rather than producing an empty file.

A13.2 The audit trail is Tourism-only

The audit trail records what everyone did, **including the museum staff it is kept on**. It is therefore exported only by the Tourism office, and is deliberately excluded from the shared export route that serves the other five reports.

A14 · Troubleshooting

A14.1 When a visitor asks for help

What their phone says, what it means, and what to do.

The visitor's screen says	What is going on	Do this
<i>Hand your admission fee to the staff at the desk...</i>	They registered and are waiting for you.	Collect the fee, Mark Paid . Their phone unlocks by itself.
<i>Show your ID at the entrance desk to claim free entry.</i>	A Baler resident waiting for an ID check.	Check the ID, Verify ID .
<i>Your group's admission has not been recorded yet.</i>	They joined a group that has not paid.	Find the group leader; Mark Paid on the group.
<i>That group code was not found for today.</i>	Wrong code, or the group was registered on another day.	Read the code from the group's row on the Front Desk page. Codes contain no O or I.
<i>Everyone in that group has already joined.</i>	The party is bigger than the headcount entered.	Correct the headcount on the group's row, then they can join.
<i>That email is already registered. Tap Sign In instead.</i>	They have visited before.	They sign in with the password they made then. There is no password reset for visitors; they may register with a different address.
<i>That email address does not look real.</i>	The part after the @ is misspelled, for example gmial.com.	Ask them to check it.
<i>Incorrect email or password. / Too many attempts.</i>	Wrong password; after six tries a one-minute pause.	Wait a minute and try again.
<i>Can't reach the museum right now.</i>	Their phone has no signal or Wi-Fi.	Point them to the museum Wi-Fi. Inside the halls the app keeps working from what it already loaded; sending feedback needs a connection.
<i>Exhibit not found: ...</i>	They typed a code that is not an active exhibit.	Check the label; the code is printed under the QR square.
<i>The camera needs a secure (https) connection.</i>	They opened the app through the wrong address.	Give them the address from the poster — it starts with https.
<i>No back camera was found / The camera is being used by another app.</i>	A problem with their phone, not the museum.	Close other camera apps, or type the exhibit code.
<i>No exhibit recognized — try moving closer</i>	Camera recognition did not match.	Scan the QR label instead. If this happens often for one exhibit, it needs more training photos (section A10).
<i>Narration file unavailable — reading the description instead</i>	No recording for that language; the phone reads the text aloud.	Nothing to do. Add narration later (section A6).
<i>Your session has expired. Please sign in again.</i>	Visitor sessions last 24 hours.	They sign in again; nothing is lost.

If a visitor asks whether the app tracks them: it records when their phone enters and leaves the museum's circle, which exhibits they opened, and what they wrote in feedback. Nothing outside the museum grounds. Nothing is sold or shared with anyone. The technical documentation sets out exactly what is collected and why, for anyone who wants the full answer.

A14.2 When the panel shows an error

Message	Meaning	Do this
<i>Page Expired / Your session expired. Reload the page and scan again.</i>	A page was left open too long.	Reload and redo the last action. Nothing was saved twice.
<i>Too many requests</i> on the AI buttons	The per-minute or daily AI budget is used up.	Wait; the daily count resets the next day.
<i>AI generation is not set up...</i>	The Google AI key is missing on the server.	Tell the maintainer. Everything else works.
<i>Gemini refused the request / declined this text</i>	Google's service or content filter.	Try again later, or reword the text.
<i>Could not create ZIP archive</i>	The server is missing its ZIP support.	Use Print All or single Download SVG ; tell the maintainer (see the technical documentation).
<i>Only today's groups can be corrected</i>	The group is from a past day.	Corrections are same-day. Tell the Tourism office.
<i>You must be at the museum to record attendance</i>	Clock-in from outside the geofence, or a vague GPS fix.	Step inside, or nearer a window, and scan again.
<i>That is not a Museo de Baler attendance code, or the code expired</i>	The kiosk code rotates every minute.	Scan the code on the kiosk screen now, not a photograph of it.
<i>Something went wrong on our side</i>	A real fault. The maintainer is emailed automatically.	Note what you were doing and the time, then try once more.
The site does not load at all	The server or the internet connection is down.	Take names on paper and enter them under Front Desk later. Visitors already inside the app keep working from what their phones hold.

A15 · Routine maintenance

A15.1 Weekly, whoever opens the museum

- Glance at the dashboard. Yesterday's visitor count should be there. **If the last few days show zero while people came**, the geofence pin or the app's address has changed — check section A9.
- Confirm the nightly backup ran: there should be two new files carrying yesterday's date in the backup folder. If there are none for two days the system emails the alert address, so make sure someone reads that inbox.

A15.2 Quarterly, the Tourism office with the maintainer

- **A restore drill.** Follow the restore procedure end to end on a scratch database and record the date. A backup nobody has ever restored is a hope, not a backup.
- **Review who holds what.** Is every account, key and credential still held by someone who works here? The access register in the repository has a table for this.
- **Reprint any QR labels that have faded** or been damaged.

The single most important item in this manual. The nightly backup is encrypted with a key kept in the server's configuration. **If that key is lost, every backup ever made with it is permanently unreadable — by anyone, including the people who wrote this system.** The key must exist in two places that cannot fail together: on the server, and somewhere that is not the server (printed and sealed in the Tourism office safe, or in the office password manager). A copy on the same machine or the same cloud account as the backups does not count. See the technical documentation.

B1 · Stack and inventory

B1.1 Platform

Component	Version and notes
Language	PHP 8.3 in production; <code>composer.json</code> requires <code>^8.2</code> . Extensions: <code>mbstring</code> <code>gd</code> <code>sodium</code> <code>pdo_mysql</code> <code>zip</code> <code>xml</code> <code>curl</code>
Framework	Laravel 12 (<code>laravel/framework ^12.0</code>), server-rendered Blade for the panel
Database	MySQL 8. Both connections pin MySQL's session time zone to <i>Asia/Manila</i>
Visitor app	Plain HTML, CSS and JavaScript — no build step and no framework . A service worker provides offline fallback; the app installs from the browser as a PWA
Panel styling	Vite, one stylesheet (<code>npm run build</code>)
Image recognition	A Teachable Machine image classifier, roughly 3 MB of static files, running in the visitor's browser
Test runner	PHPUnit 11 on in-memory SQLite; <code>node --test</code> for the service worker
Node	20 or newer (24 on the development machine)

Production dependencies, and what each is for

Package	Why it is here
<code>laravel/framework ^12.0</code>	The application framework
<code>chillerlan/php-qrcode ^6.0</code>	QR labels for the display cases, as SVG, and the rotating staff-attendance kiosk code
<code>mpdf/mpdf ^8.3</code>	PDF reports. Chosen over lighter converters because its HTML header support repeats the museum letterhead on page two onward — a letterhead on only the first page of a twelve-page DTR is not a letterhead
<code>phpoffice/phpspreadsheet ^5.10</code>	XLSX exports (logbook, DTR, visitors)
<code>phpoffice/phpword ^1.4</code>	DOCX exports (exhibits, feedback)
<code>laravel/tinker</code>	REPL for maintenance

There is **no** queue worker, websocket server, Redis instance or search engine to keep alive. The database is the cache, session and queue driver. This is a deliberate operational choice for a municipal museum with no full-time systems administrator.

B1.2 Code inventory

Counted from the working tree at the time of writing.

Item	Count
Registered routes (total)	119
of which the visitor API (<code>/api/v1</code>)	16
of which the panel and framework endpoints	103
Panel controllers	18
API controllers	10

Item	Count
Middleware classes	11
Eloquent models	23
Migrations	51
Database tables (23 domain + 8 framework)	31
Artisan console commands (project-specific)	7
PHPUnit tests / assertions, all passing	411 / 4,913
Service-worker tests, all passing	15

B1.3 What is authored and what is generated

A frequent question, because the folder on disk is far larger than the project. Everything not checked into version control is either generated by a tool or uploaded by a user, and all of it is listed in `.gitignore`.

On disk	Size	In the repo?	Where it comes from
<code>app/ routes/ resources/ config/ database/ tests/ docs/ public/visitor/</code>	~4 MB	Yes	Written by the team — this is the project
<code>vendor/</code>	170 MB	No	<code>composer install</code> reads <code>composer.lock</code>
<code>node_modules/</code>	36 MB	No	<code>npm ci</code> reads <code>package-lock.json</code>
<code>public/images/ public/audio/</code>	103 MB	No	Uploaded exhibit photos, training photos, narration
<code>storage/app/backups/</code>	~108 MB	No	<code>php artisan db:backup</code> , pruned automatically
<code>storage/logs/ storage/framework/</code>	~11 MB	No	Runtime logs and compiled Blade cache
<code>public/build/</code>	~1 MB	No	<code>npm run build</code>

The two lock files are the point: `composer.lock` and `package-lock.json` are committed and pin every dependency to an exact version. That is why `vendor/` need not be — any machine rebuilds it identically, and continuous integration does exactly that on every push before running the suite.